# Foundations of Programming

## Assignment No. : 06

**PROBLEM STATEMENT:**Write a C function to compute the factorial of a number with and without recursion.

**OBJECTIVE:**

To demonstrate two different approaches for calculating the factorial of a number: **iterative (non-recursive)** and **recursive**, showcasing the versatility of problem-solving methods in programming.

**THEORY:**

**1) Explain the difference between the iterative and recursive approaches to calculating the factorial of a number.**Iterative uses loops to compute factorial step-by-step, while recursive calls the same function repeatedly using .Iterative is faster and memory-efficient, whereas recursion is more elegant but uses extra stack memory.

**2) What is the base case in the recursive function for computing factorials, and why is it necessary?**The base case is (or ), which stops further recursive calls.It is necessary to prevent infinite recursion and provide a terminating condition.

**3) What is a function and why is it important?**A function is a reusable block of code designed to perform a specific task.It improves modularity, reduces repetition, and makes programs easier to read, test, and maintain.

**4) How does the program handle invalid input, such as negative numbers, and why is this important?**The program checks if the number is negative and displays an error message or stops execution.This is important because factorial is not defined for negative numbers and prevents incorrect results.

**5) Why is handling invalid input an important programming concept?**Handling invalid input ensures correct logic, prevents errors, and improves program reliability.It helps in structured problem solving and makes programs easier to maintain and debug.

**PLATFORM:** Linux

**INPUT:**

```
#include <stdio.h>

int main()

{

   int n, i;

   long long fact = 1;

   printf("Enter a number: ");

   scanf("%d", &n);

   for (i = 1; i <= n; i++)
```

```
    {
        fact = fact * i;
    }
    printf("Factorial = %lld\n", fact);
    return 0;
}
```

```c
#include <stdio.h>

// iterative and recursive
long long factorialIterative(int n) {
    long long result = 1;
    int i;
    for (i = 1; i <= n; i++)
        result *= i;
    return result;
}

long long factorialRecursive(int n) {
    if (n <= 1)
        return 1;
    return n * factorialRecursive(n - 1);
}

int main() {
    int number;

    printf("Enter a number: ");
    scanf("%d", &number);

    if (number < 0) {
        printf("Factorial is not defined for negative numbers.\n");
        return 1;
    }

    printf("Factorial (Iterative): %lld\n", factorialIterative(number));
    printf("Factorial (Recursive): %lld\n", factorialRecursive(number));

    // factorial computation methods
    return 0;
}
```

**OUTPUT:**

Enter a number: 4

Factorial = 24

**CONCLUSION:** Thus, we have successfully executed factorial of a number with and without recursion.

**FAQs:**

1. What is the purpose of calculating factorial in programming?

2. What is the difference between recursive and non-recursive (iterative) methods?